

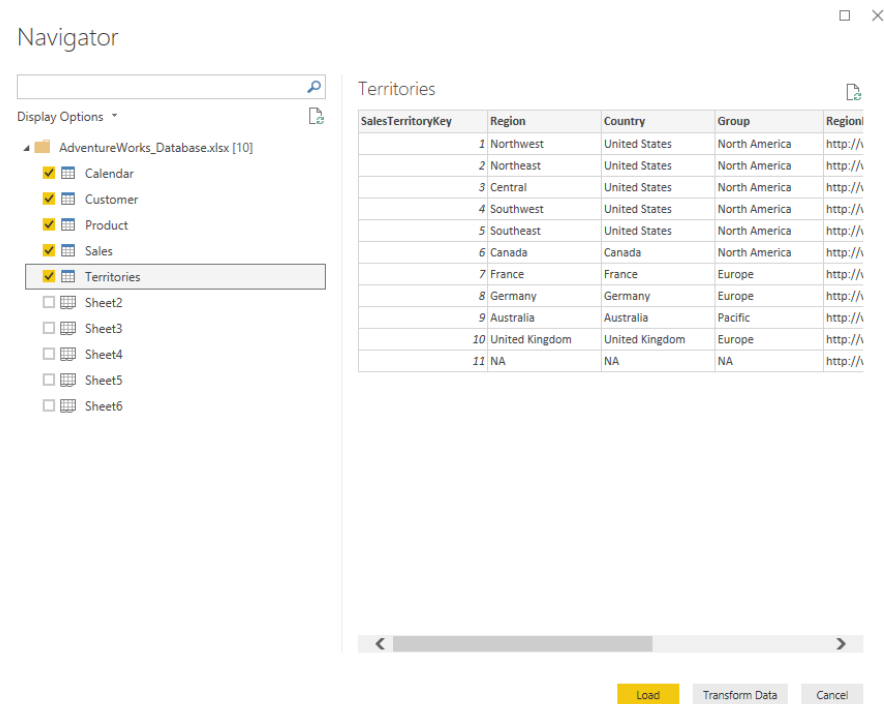
Getting your dataset ready for Data Analytics

Introduction

In this workbook, you will work with a new Database (AdventureWorksDW2017) to do Data Analytics in Power BI. Before doing so, we need to cleanse the data and create the correct relationships between tables. Let's get started!

Step 1: Get Data and Filtering Values

For this workbook, you will need to download a new file from Blackboard (AdventureWorks_Database). Use the Get Data -> Excel File button to connect to the file. You should now see the window on the right, which shows which data is available in that file. Select the top 5 tables and click load.



Navigator

Display Options ▾

AdventureWorks_Database.xlsx [10]

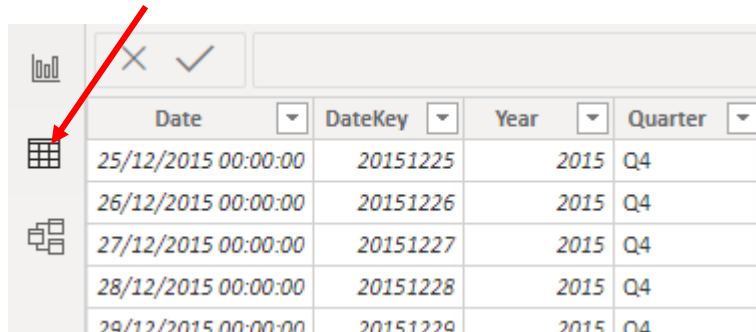
- ☒ Calendar
- ☒ Customer
- ☒ Product
- ☒ Sales
- ☒ Territories
- ☐ Sheet2
- ☐ Sheet3
- ☐ Sheet4
- ☐ Sheet5
- ☐ Sheet6

Territories

SalesTerritoryKey	Region	Country	Group	Region1
1	Northwest	United States	North America	http://
2	Northeast	United States	North America	http://
3	Central	United States	North America	http://
4	Southwest	United States	North America	http://
5	Southeast	United States	North America	http://
6	Canada	Canada	North America	http://
7	France	France	Europe	http://
8	Germany	Germany	Europe	http://
9	Australia	Australia	Pacific	http://
10	United Kingdom	United Kingdom	Europe	http://
11	NA	NA	NA	http://

Load Transform Data Cancel

Go to the Data tab (the middle one on the left) and check that your data is displayed correctly. Now save the file as “AdventureWorksTask”. You have just uploaded five tables. Let’s have a look at each of the tables we have just uploaded.

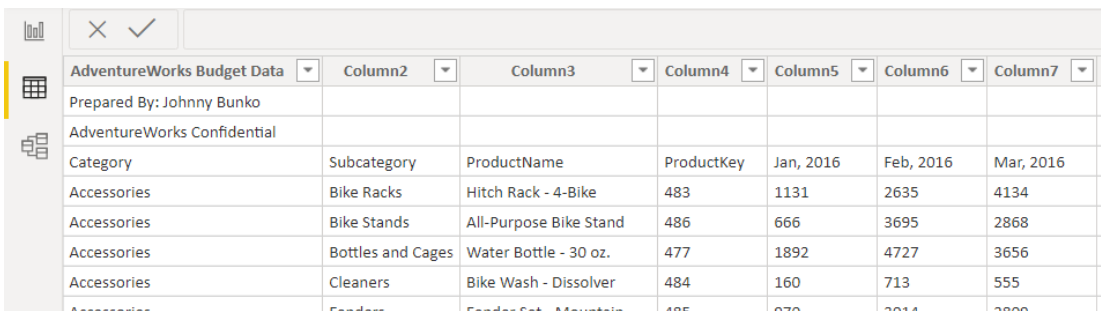


Date	DateKey	Year	Quarter
25/12/2015 00:00:00	20151225	2015	Q4
26/12/2015 00:00:00	20151226	2015	Q4
27/12/2015 00:00:00	20151227	2015	Q4
28/12/2015 00:00:00	20151228	2015	Q4
29/12/2015 00:00:00	20151229	2015	Q4

- **Calendar.** This table shows information about each date (e.g. day, month, year, day of the week, quarter, etc.). This is going to be useful when we need to filter the transactions based on a specific period of time.
- **Customer.** This table stores the information about customers (e.g. name, surname, age, birthdate, etc.)
- **Product.** This table reports the information about the products available (e.g., the product name, category, model name, etc.)
- **Sales.** This is the most important table of the model. It reports the details of each transaction (e.g. product sold, sales amount, ship date, etc.)
- **Territories.** This table reports the details of territories where the products have been sold (e.g. region, country, etc.).

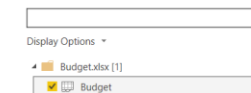
Before starting the data pre-cleansing process, use the Get Data button (Excel file) to upload the **Budget** Excel File (you can find this file on Blackboard). Tick the box next to the table name, then select Load to load this table in your Power BI project.

This table reports the budget for each month in 2016 for different products.



AdventureWorks Budget Data	Column2	Column3	Column4	Column5	Column6	Column7
Prepared By: Johnny Bunko						
AdventureWorks Confidential						
Category	Subcategory	ProductName	ProductKey	Jan, 2016	Feb, 2016	Mar, 2016
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	1131	2635	4134
Accessories	Bike Stands	All-Purpose Bike Stand	486	666	3695	2868
Accessories	Bottles and Cages	Water Bottle - 30 oz.	477	1892	4727	3656
Accessories	Cleaners	Bike Wash - Dissolver	484	160	713	555
Accessories	Fenders	Fender Set - Mountain	485	970	2014	1908

Navigator

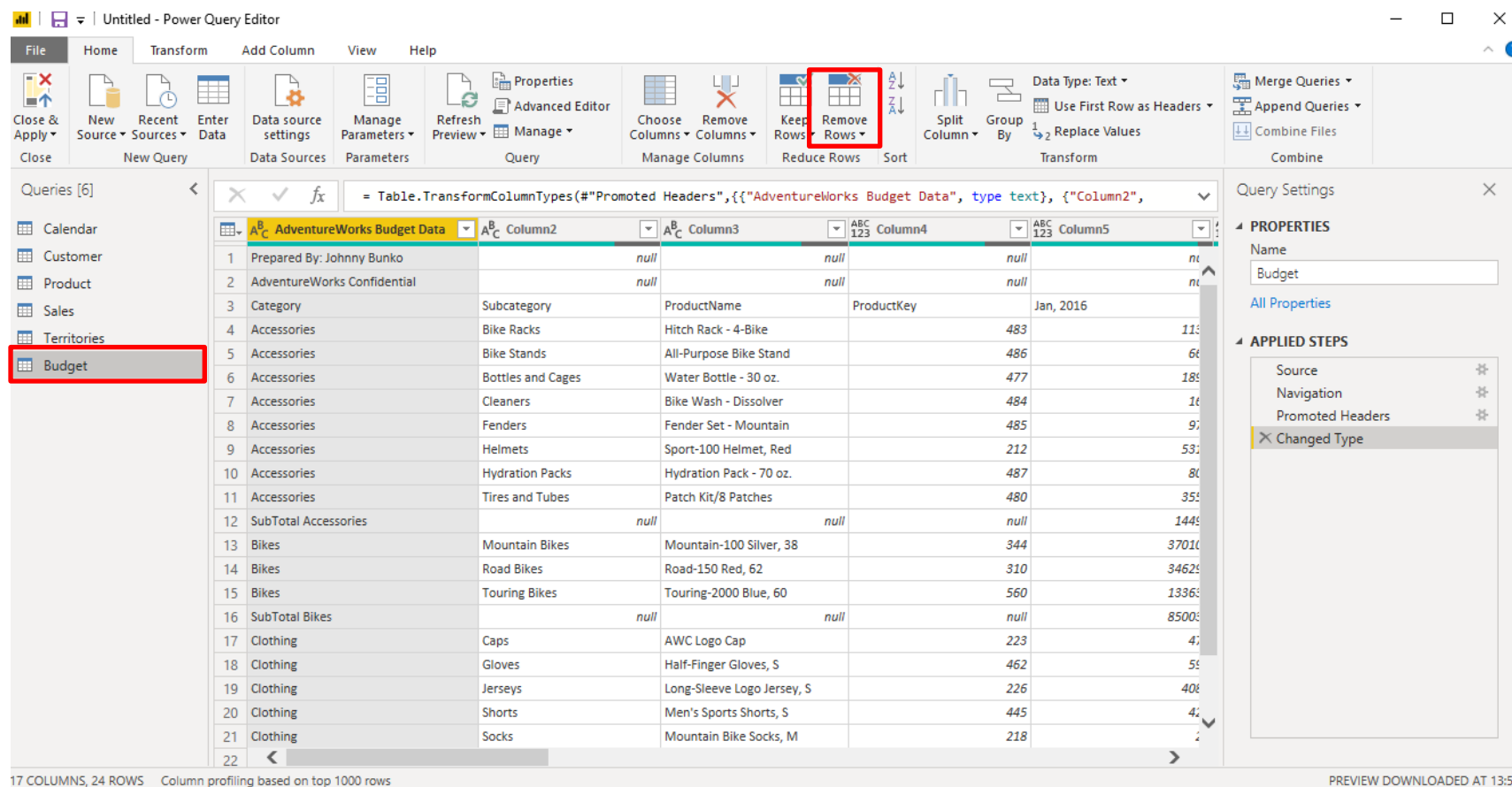


Budget

AdventureWorks Budget Data	Column2	Column3	Column4
Prepared By: Johnny Bunko			
AdventureWorks Confidential			
Category	Subcategory	ProductName	ProductKey
Accessories	Bike Racks	Hitch Rack - 4-Bike	483
Accessories	Bike Stands	All-Purpose Bike Stand	486
Accessories	Bottles and Cages	Water Bottle - 30 oz.	477
Accessories	Cleaners	Bike Wash - Dissolver	484
Accessories	Fenders	Fender Set - Mountain	485
Accessories	Helmets	Sport-100 Helmet, Red	487
Accessories	Hydration Packs	Hydration Pack - 70 oz.	488
Accessories	Tires and Tubes	Patch Kit/8 Patches	489
SubTotal Accessories			
Bikes	Mountain Bikes	Mountain-100 Silver, 38	501
Bikes	Road Bikes	Road-150 Red, 62	502
Bikes	Touring Bikes	Touring-2000 Blue, 60	503
SubTotal Bikes			
Clothing	Caps	AWC Logo Cap	504
Clothing	Gloves	Half-Finger Gloves, S	505
Clothing	Jerseys	Long-Sleeve Logo Jersey, S	506
Clothing	Shorts	Men's Sports Shorts, S	507
Clothing	Socks	Mountain Bike Socks, M	508
Clothing	Vests	Classic Vest, S	509
SubTotal Clothing			

Load Transform Data Cancel

Once you have uploaded all the files, we can start transforming the Data. Select Transform Data (or Edit Queries) from the Home ribbon to open the Power Query Editor. This will display the window below. Select the **Budget** table, from the left menu.



Queries [6]

- Calendar
- Customer
- Product
- Sales
- Territories
- Budget**

Table.TransformColumnTypes(#"Promoted Headers",{{"AdventureWorks Budget Data", type text}, {"Column2", type text}, {"Column3", type text}, {"Column4", type text}, {"Column5", type text}})

	Column2	Column3	Column4	Column5
1	Prepared By: Johnny Bunko	null	null	null
2	AdventureWorks Confidential	null	null	null
3	Category	Subcategory	ProductName	ProductKey
4	Accessories	Bike Racks	Hitch Rack - 4-Bike	483
5	Accessories	Bike Stands	All-Purpose Bike Stand	486
6	Accessories	Bottles and Cages	Water Bottle - 30 oz.	477
7	Accessories	Cleaners	Bike Wash - Dissolver	484
8	Accessories	Fenders	Fender Set - Mountain	485
9	Accessories	Helmets	Sport-100 Helmet, Red	212
10	Accessories	Hydration Packs	Hydration Pack - 70 oz.	487
11	Accessories	Tires and Tubes	Patch Kit/8 Patches	480
12	SubTotal Accessories	null	null	null
13	Bikes	Mountain Bikes	Mountain-100 Silver, 38	344
14	Bikes	Road Bikes	Road-150 Red, 62	310
15	Bikes	Touring Bikes	Touring-2000 Blue, 60	560
16	SubTotal Bikes	null	null	null
17	Clothing	Caps	AWC Logo Cap	223
18	Clothing	Gloves	Half-Finger Gloves, S	462
19	Clothing	Jerseys	Long-Sleeve Logo Jersey, S	226
20	Clothing	Shorts	Men's Sports Shorts, S	445
21	Clothing	Socks	Mountain Bike Socks, M	218
22				

17 COLUMNS, 24 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:57

Query Settings

PROPERTIES

Name: Budget

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type**

We need to remove the top rows with null values, so click on the **Remove Rows** button at the top, then select **Remove top rows**. Set the number of rows to be deleted equal to **2** (or 3 depending on how many top rows are showing null values in your file), as shown below. Then click ok.

1	Prepared By: Johnny Bunko	Column2	Column3	Column4	Column5
2	AdventureWorks Confidential	Column2	Column3	Column4	Column5
3	Category	Subcategory	ProductName	ProductKey	Jan, 2016
4	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	1131
5					666
6					1892
7					160
8					970
9					5317
10					809
11					3554
12					14499
13					370105
14					346295
15					133631

I need to remove these two rows

Remove Top Rows

Specify how many rows to remove from the top.

Number of rows

2

OK

Cancel

A new step has been added on the applied steps pane, on the right. Now let's make the first row the headers. Click on use first row as headers to promote the first row as headers.

Untitled - Power Query Editor

File Home Transform Add Column View Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [6]

- Calendar
- Customer
- Product
- Sales
- Territories
- Budget

Table.Skip(#"Changed Type",2)

1	Category	Subcategory	ProductName	ProductKey	Jan, 2016
2	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	1131
3	Accessories	Bike Stands	All-Purpose Bike Stand	486	666
4	Accessories	Bottles and Cages	Water Bottle - 30 oz.	477	1892
5	Accessories	Cleaners	Bike Wash - Dissolver	484	160
6	Accessories	Fenders	Fender Set - Mountain	485	970
7	Accessories	Helmets	Sport-100 Helmet, Red	212	5317
8	Accessories	Hydration Packs	Hydration Pack - 70 oz.	487	809
9	Accessories	Tires and Tubes	Patch Kit/8 Patches	480	3554
10	SubTotal Accessories				14499

Query Settings

PROPERTIES

Name

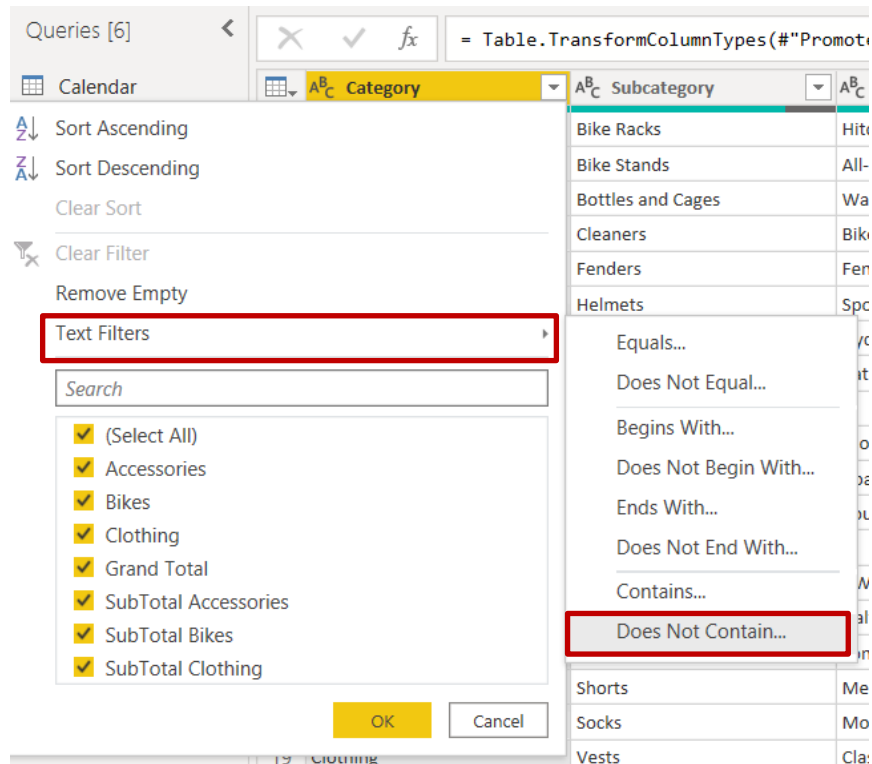
Budget

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Top Rows

The next step is to filter out the *null* values. Before doing so, note that the null values are all on the rows corresponding to **SubTotal** and **GrandTotal** rows. Hence, we want to remove all the rows that contain the “SubTotal” word. To do so, click on the drop-down menu in the column header of the first column (Category) and select Text-Filters -> does not contain... as shown below:



We want to **leave (or filter)** the rows that do not contain the word “Total”(uppercase), so write Total in the Filter Rows window and click OK (see below). Make sure you write “Total” in upper case, otherwise it will not work. All the SubTotal columns have disappeared.

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'Accessories'

does not contain

☒ And ☐ Or

OK

Cancel

The new table should look like the one below. The rows containing null values are not there anymore.

Queries [6]		= Table.SelectRows("#Filtered Rows", each not Text.Contains([Category], "Total"))					
	Category	Subcategory	ProductName	ProductKey	Jan, 2016	Feb, 2016	
1	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	1131		
2	Accessories	Bike Stands	All-Purpose Bike Stand	486	666		
3	Accessories	Bottles and Cages	Water Bottle - 30 oz.	477	1892		
4	Accessories	Cleaners	Bike Wash - Dissolver	484	160		
5	Accessories	Fenders	Fender Set - Mountain	485	970		
6	Accessories	Helmets	Sport-100 Helmet, Red	212	5317		
7	Accessories	Hydration Packs	Hydration Pack - 70 oz.	487	809		
8	Accessories	Tires and Tubes	Patch Kit/8 Patches	480	3554		
9	Bikes	Mountain Bikes	Mountain-100 Silver, 38	344	370105		
10	Bikes	Road Bikes	Road-150 Red, 62	310	346295		
11	Bikes	Touring Bikes	Touring-2000 Blue, 60	560	133631		
12	Clothing	Caps	AWC Logo Cap	223	479		
13	Clothing	Gloves	Half-Finger Gloves, S	462	598		
14	Clothing	Jerseys	Long-Sleeve Logo Jersey, S	226	4087		
15	Clothing	Shorts	Men's Sports Shorts, S	445	421		
16	Clothing	Socks	Mountain Bike Socks, M	218	24		
17	Clothing	Vests	Classic Vest, S	471	980		

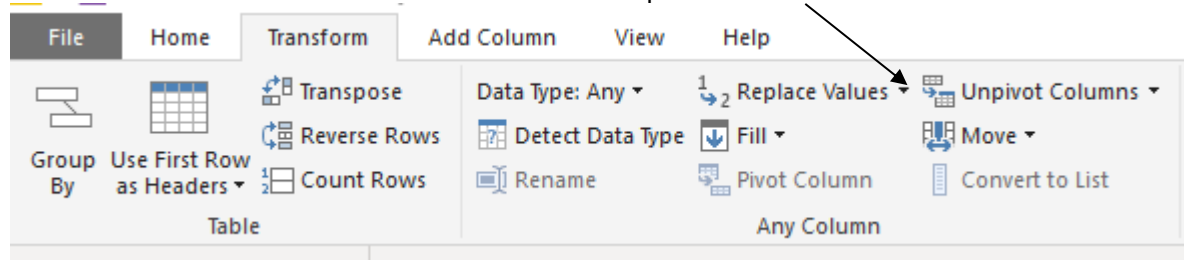
Step 2: Power Pivot

Now that our data is ready, we need to make it easier to read and work with. We are going to change columns to rows. At the moment, the budget table has a column for each month. In this step, we are going to merge all the months columns into one using Power Pivot. In the Budget table, select all the columns from January 2016 to December 2016 (you can do this by clicking on the first column (Jan, 2016), press Shift on your keyboard and click on the last column (Dec, 2016)). You should have all the date columns highlighted as shown below.

Queries [6] ✕ ✓ fx = Table.SelectRows(#"Filtered Rows", each not Text.Contains([Category], "Total"))

	Jan, 2016	Feb, 2016	Mar, 2016	Apr, 2016	May, 2016	Jun, 2016
1	1131	2635	4134	2179	2637	32
2	666	3695	2868	4862	3439	46
3	1892	4727	3656	4449	4051	62
4	160	713	555	656	369	5
5	970	3014	2809	4259	3638	37
6	5317	16221	16752	16552	17204	253
7	809	2684	2917	3425	2716	32
8	3554	18758	20905	18046	21680	224
9	370105	326786	384811	439822	458523	6194
10	346295	289524	355097	346783	399691	5460
11	133631	165941	178287	265901	286630	4452

Then click on the Transform ribbon and select Unpivot Columns

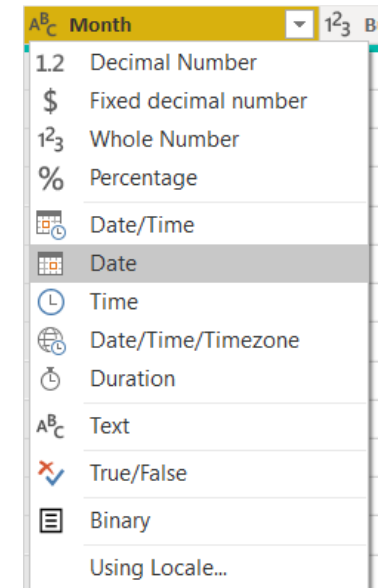


The dates have now been merged into one single column (Attribute) and the subtotal for that year has been copied in the Value column. **Rename the Attribute** column as **Month** and the **Value** column as **Budget Amount** (you can do this by right-clicking on the column-header and selecting Rename, make sure you select only one column before right-clicking, otherwise the option “Rename” will not be displayed). The final result is shown below.

fx = Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Month"}, {"Value", "Budget Amount"}})

	ABC ProductName	123 ProductKey	123 Grand Total	ABC Month	123 Budget Amount
1	Hitch Rack - 4-Bike	483	36861	Jan, 2016	1131
2	Hitch Rack - 4-Bike	483	36861	Feb, 2016	2635
3	Hitch Rack - 4-Bike	483	36861	Mar, 2016	4134
4	Hitch Rack - 4-Bike	483	36861	Apr, 2016	2179
5	Hitch Rack - 4-Bike	483	36861	May, 2016	2637
6	Hitch Rack - 4-Bike	483	36861	Jun, 2016	3279
7	Hitch Rack - 4-Bike	483	36861	Jul, 2016	2218
8	Hitch Rack - 4-Bike	483	36861	Aug, 2016	3287
9	Hitch Rack - 4-Bike	483	36861	Sep, 2016	3885
10	Hitch Rack - 4-Bike	483	36861	Oct, 2016	2484
11	Hitch Rack - 4-Bike	483	36861	Nov, 2016	5441
12	Hitch Rack - 4-Bike	483	36861	Dec, 2016	3551
13	All-Purpose Bike Stand	486	38757	Jan, 2016	666
14	All-Purpose Bike Stand	486	38757	Feb, 2016	3695
15	All-Purpose Bike Stand	486	38757	Mar, 2016	2868
16	All-Purpose Bike Stand	486	38757	Apr, 2016	4862
17	All-Purpose Bike Stand	486	38757	May, 2016	3439

Change the data-type of the Month column to Date. For this, you can click on the ABC symbol in the column header and select “Date”. The final table is shown below.



	Category	Subcategory	ProductName	ProductKey	Grand Total	Month	Budget Amount
1	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/01/2016	1131
2	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/02/2016	2635
3	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/03/2016	4134
4	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/04/2016	2179
5	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/05/2016	2637
6	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/06/2016	3279
7	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/07/2016	2218
8	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/08/2016	3287
9	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/09/2016	3885
10	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/10/2016	2484
11	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/11/2016	5441
12	Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	01/12/2016	3551
13	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/01/2016	666
14	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/02/2016	3695
15	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/03/2016	2868
16	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/04/2016	4862
17	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/05/2016	3439
18	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/06/2016	4612
19	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/07/2016	2774
20	Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	01/08/2016	3003

Now you are ready to click on File -> Close&Apply to apply these changes and go back to main Power BI window.

Step 3: Working with Relationships

In this task, we want to compare information in the Sales and in the Budget tables. To do so, both tables need to be “connected” to the same Calendar table. This means that we have to create a new relationship between the Sales and Calendar tables and between the Budget and Calendar tables.

In the left side of the main Power BI window, select the Model View.

Delete all the relationships between the tables by right-clicking on them and select Delete. You can watch this short video to find out how to delete all the relationships at once: [link to video](#). Rearrange your tables as shown below

You can move tables around by clicking on the header of the table. Drop them where you want.



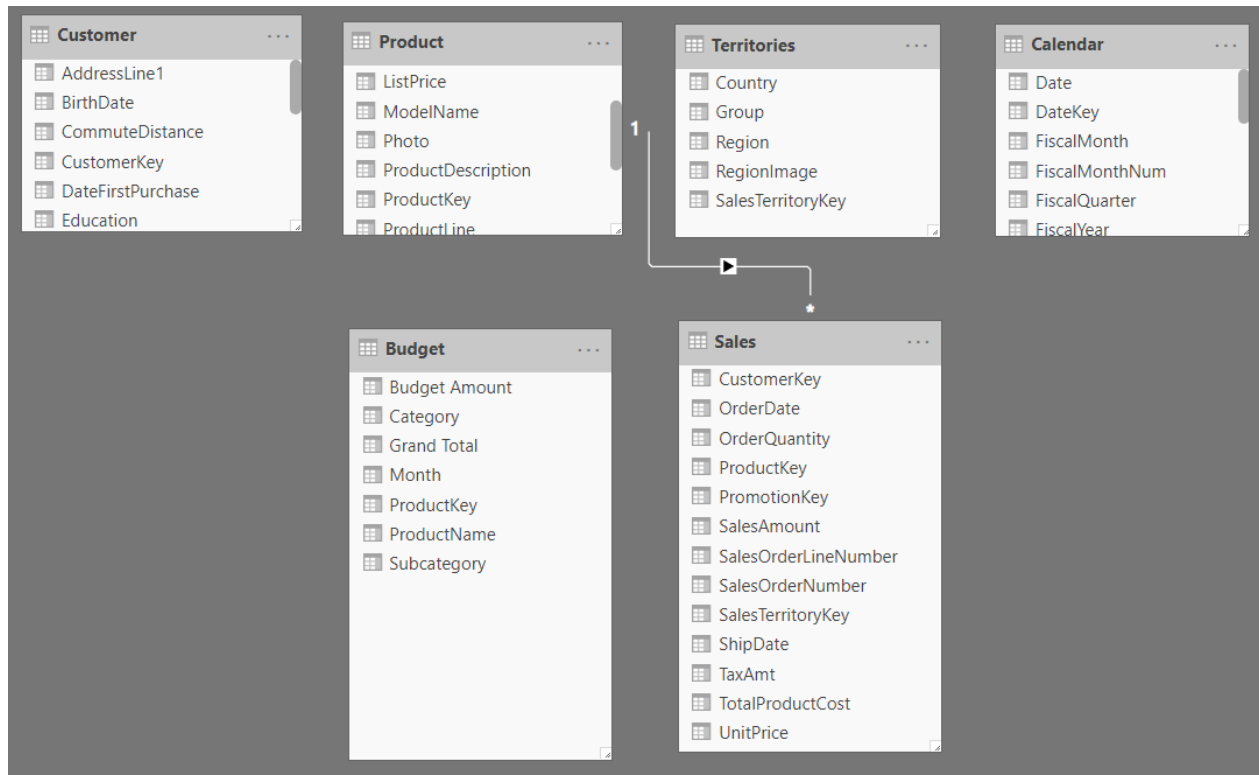
Customer	Product	Territories	Calendar
AddressLine1	Category	Country	Date
BirthDate	Color	Group	DateKey
CommuteDistance	DaysToManufacture	Region	FiscalMonth
CustomerKey	ListPrice	RegionImage	FiscalMonthNum
DateFirstPurchase	ModelName	SalesTerritoryKey	FiscalQuarter

Sales	Budget
CustomerKey	Category
OrderDate	Date
OrderQuantity	Grand Total
ProductKey	ProductKey
PromotionKey	ProductName
SalesAmount	Subcategory
SalesOrderLineNumber	Total Budget
SalesOrderNumber	Budget
SalesTerritoryKey	
ShipDate	
TaxAmt	
Total Sales Amount	
TotalProductCost	

If you are familiar with data modelling, you might have noticed that I have put the Lookup tables (the who, what, when, where and how) at the top and the Data Tables (which record transactions and have many columns) at the bottom.

Please note that in this example we have more than one fact table. This can happen if the model integrates data from different sources.

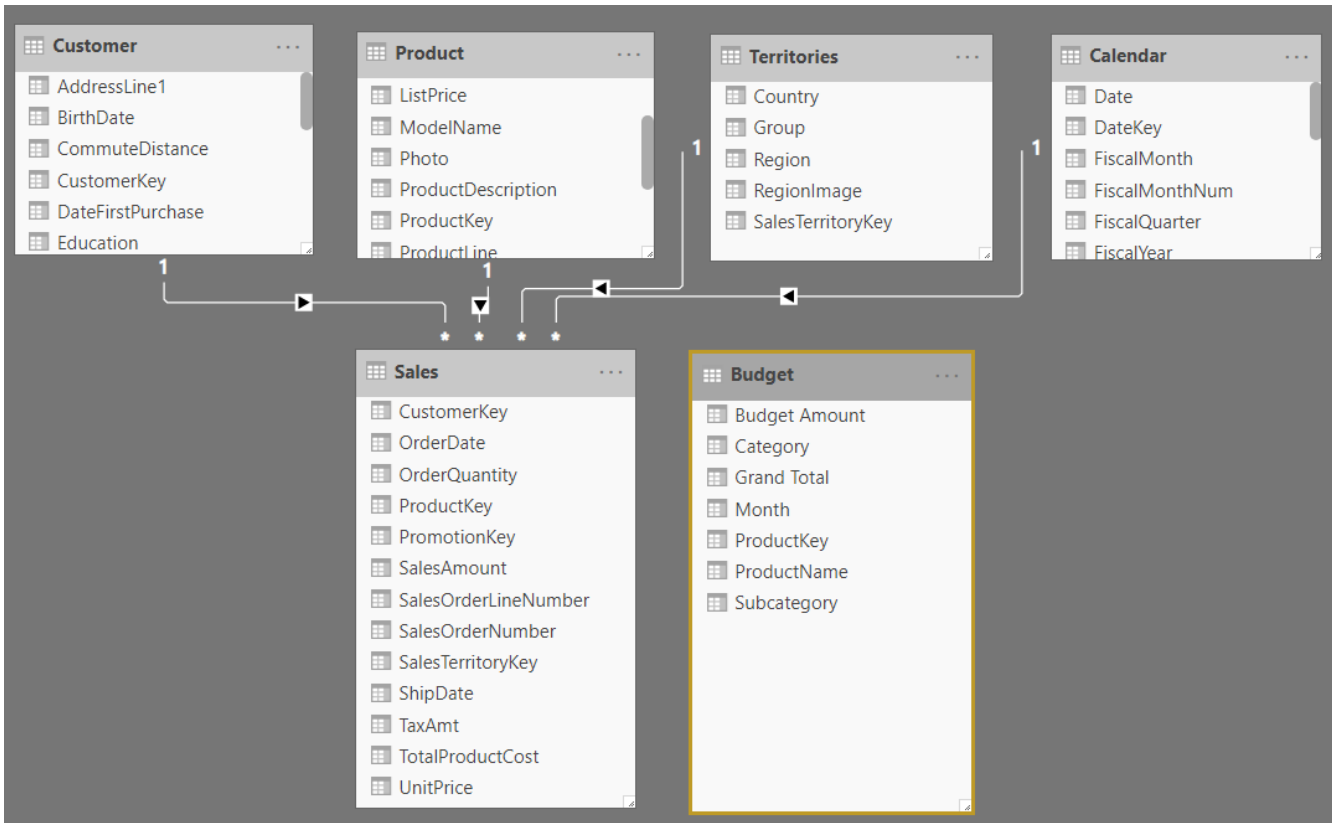
It's now time to create the model relationships. In Power BI, creating relationships is really easy. The first relationship will be between the **Sales** and **Product** tables based on the "**Product Key**" column. Look for "**Product Key**" in the **Sales** table, click on it and drag and drop it in the **Product** table exactly on the "**Product Key**" column. A new relationship has just been added to your model!



Repeat the same task for the following columns:

- **CustomerKey** between the **Customer** and the **Sales** tables
- **SalesTerritoryKey** between the **Territories** and the **Sales** tables
- **OrderDate** in the **Sales** table has to be connected to the **Date** in the **Calendar** table. (Drop OrderDate on Date).

Your model so far should look like the one below:



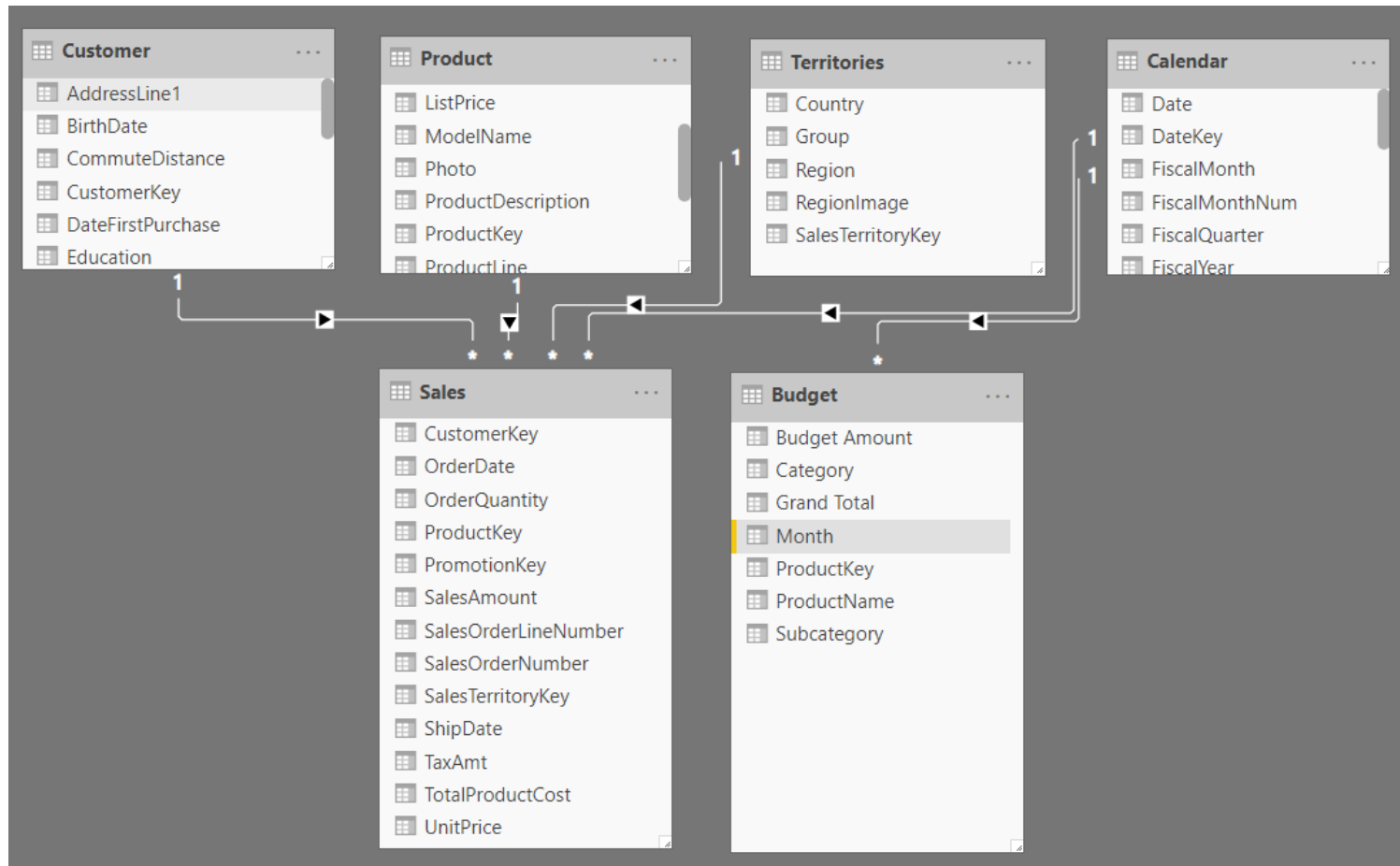
If you look at the direction of the relationships, you will notice that the relationships are filtering from the look-up tables towards the Sales and Budget tables.

This is because we want to filter the Sales or Budget information based on the information stored in the look-up tables (e.g. Customer or Product).

The next step is to create relationships between the Budget table and the LookUp tables. As you did before, create the following relationship:

- Connect the **Month** in the **Budget** table to **Date** in the **Calendar** table

Before moving to the next section, make sure your relationships look like the ones below.



Remember that the calendar table is one of the most important table in your model, and each model should have one. Notice that the **Sales** and **Budget** tables are “indirectly” connected to each other through the Calendar table. This is a really useful feature when creating reports as we will see in the next tutorial.

Make sure you save this file as “AdventureWorksTask”. In the next task, you will use this Power BI project to perform Data Analytics using DAX.